

# Laboratory Record

submitted as part of

## Programming for Data Science Lab (MACSE502)

### LAB ASSESEMENT-1

by



Reg. No: 26MML0045

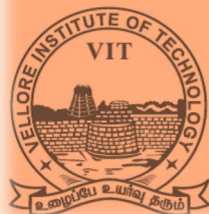
VEDANT NIMKAR

To



Dr M Rajasekhara Babu

School of Computer Science and Engineering



# VIT<sup>®</sup>

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

JULY 2026

**2026 FALL SEMESTER**  
**Programming for Data Science Lab (MACSE502) Record**  
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| Week No | Date       | Sl. No. | Type | Topic  | Title with download Link for document               | Problem Statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Page No. |
|---------|------------|---------|------|--------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1       | 09-07-2026 | 1       | PP   | Pandas | <a href="#">Basic Data Manipulation with Pandas</a> | Create a DataFrame with columns Name, Age, and City containing data for five individuals. Perform the following operations: select only the Name and Age columns, filter rows where Age is greater than 25, add a new column Country with a default value, and sort the DataFrame by Age in descending order. Finally, calculate the average age of the individuals.                                                                                                                | 3        |
| 1       | 09-07-2026 | 2       | EP   | Pandas | <a href="#">Product Sales Analysis Using Pandas</a> | An e-commerce company maintains product sales records containing ProductID, SalesAmount, and Category. Create a DataFrame for 20 products and perform the following operations: select only the ProductID and SalesAmount columns, filter the products whose SalesAmount is greater than Rs. 50,000, add a new column Region with the default value "South India", sort the products by SalesAmount in descending order, and calculate the average SalesAmount of all the products. | 9        |

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**Practice Program/ Exercise Program**

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| Title                   | Basic Data Manipulation with Pandas                                                                                                                                                                                                                                                                                                                                                                                                              |                      |         |                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------|---------------------------------------------------------------|
| Problem Statement       | Create a DataFrame with columns Name, Age, and City containing data for five individuals. Perform the following operations: select only the Name and Age columns, filter rows where Age is greater than 25, add a new column Country with a default value, and sort the DataFrame by Age in descending order. Finally, calculate the average age of the individuals.                                                                             |                      |         |                                                               |
| Objectives              | <ul style="list-style-type: none"><li>• To understand how to create a Pandas DataFrame from a Python dictionary.</li><li>• To learn column selection techniques using Pandas.</li><li>• To apply filtering conditions on DataFrame rows.</li><li>• To create and populate new columns in a DataFrame.</li><li>• To sort data based on column values.</li><li>• To perform basic statistical analysis using Pandas aggregate functions.</li></ul> |                      |         |                                                               |
| Dataset URL Link / File | The dataset is created manually within the program using a Python dictionary. It contains the details of five individuals as shown below.                                                                                                                                                                                                                                                                                                        |                      |         |                                                               |
|                         | Name                                                                                                                                                                                                                                                                                                                                                                                                                                             |                      | Age     | City                                                          |
|                         | Alice                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      | 23      | New York                                                      |
|                         | Bob                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      | 30      | London                                                        |
|                         | Charlie                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      | 28      | Paris                                                         |
|                         | David                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      | 35      | Tokyo                                                         |
|                         | Emma                                                                                                                                                                                                                                                                                                                                                                                                                                             |                      | 22      | Sydney                                                        |
|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                      |         |                                                               |
| Functions Details       | Sl. No.                                                                                                                                                                                                                                                                                                                                                                                                                                          | Function             | Package | Description of function                                       |
|                         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                | DataFrame()          | pandas  | Creates a DataFrame from structured data such as a dictionary |
|                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                | df[['col1','col 2']] | pandas  | Selects a specific subset of columns from a DataFrame         |
|                         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                | df[condition]        | pandas  | Filters DataFrame rows using a                                |

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|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                 |                                                       |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------|-------------------------------------------------------|
|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                 | Boolean condition                                     |
|                                                                                   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | sort_values() | pandas          | Sorts DataFrame rows based on the values of a column  |
|                                                                                   | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | mean()        | pandas          | Calculates the arithmetic average of a numeric column |
|                                                                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | print()       | Python built-in | Displays the intermediate and final results           |
|                                                                                   | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |                 |                                                       |
|                                                                                   | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |                 |                                                       |
|                                                                                   | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |                 |                                                       |
|                                                                                   | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |                 |                                                       |
|                                                                                   | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |                 |                                                       |
|                                                                                   | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |                 |                                                       |
| <b>Steps/<br/>Process/<br/>Procedure/<br/>Algorithm/<br/>Flowchart/<br/>Model</b> | <b>Step 1</b><br>Import the Pandas library and create a dictionary containing Name, Age, and City data.<br><b>Step 2</b><br>Convert the dictionary into a Pandas DataFrame.<br><b>Step 3</b><br>Select only the Name and Age columns from the DataFrame.<br><b>Step 4</b><br>Filter the DataFrame to display individuals whose age is greater than 25.<br><b>Step 5</b><br>Add a new column named Country and assign a default value of India.<br><b>Step 6</b><br>Sort the DataFrame in descending order based on the Age column.<br><b>Step 7</b><br>Calculate the average age of all individuals using the mean() function.<br><b>Step 8</b><br>Display all intermediate and final results. |               |                 |                                                       |
|                                                                                   | <b>Program</b> <pre># Load Necessary Packages import pandas as pd  # Step 1: Create dataset data = {     'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Emma'],     'Age': [23, 30, 28, 35, 22],     'City': ['New York', 'London', 'Paris', 'Tokyo', 'Sydney']}  # Step 2: Create DataFrame df = pd.DataFrame(data)  print("Original DataFrame") print(df)</pre>                                                                                                                                                                                                                                                                                                                               |               |                 |                                                       |

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```
# Step 3: Select Name and Age columns
name_age = df[['Name', 'Age']]
print("\nName and Age")
print(name_age)

# Step 4: Filter Age > 25
filtered_df = df[df['Age'] > 25]
print("\nAge > 25")
print(filtered_df)

# Step 5: Add Country column
df['Country'] = 'India'
print("\nCountry Added")
print(df)

# Step 6: Sort by Age in descending order
sorted_df = df.sort_values(by='Age', ascending=False)
print("\nSorted Data")
print(sorted_df)

# Step 7: Calculate Average Age
average_age = df['Age'].mean()
print("\nAverage Age =", average_age)
```

**Output  
Snapshots  
With  
Explanation/  
Observations**

Original DataFrame

|   | Name    | Age | City     |
|---|---------|-----|----------|
| 0 | Alice   | 23  | New York |
| 1 | Bob     | 30  | London   |
| 2 | Charlie | 28  | Paris    |
| 3 | David   | 35  | Tokyo    |
| 4 | Emma    | 22  | Sydney   |

**Figure 1.1: Original DataFrame**

Figure 1.1 displays the original DataFrame created from a Python dictionary containing information about five individuals. The DataFrame consists of three columns, namely Name, Age, and City, where each row represents a different individual and the corresponding details. The `pd.DataFrame()` function converts the dictionary data into a structured tabular format, which makes it easier to store, view, and manipulate the data using Pandas. This DataFrame serves as the starting dataset for all the subsequent operations such as column selection, filtering, adding new columns, sorting, and calculating statistical measures.

| Name and Age |         |     |
|--------------|---------|-----|
|              | Name    | Age |
| 0            | Alice   | 23  |
| 1            | Bob     | 30  |
| 2            | Charlie | 28  |
| 3            | David   | 35  |
| 4            | Emma    | 22  |

**Figure 1.2: Selected Name and Age Columns**

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Figure 1.2 shows the DataFrame after selecting only the Name and Age columns from the original dataset using Pandas column indexing. This operation extracts the required columns and creates a new DataFrame named `name_age`, removing the City column from the view. Such column selection helps to focus on the relevant attributes for the analysis and reduces unnecessary data processing.

|   | Age > 25               |
|---|------------------------|
|   | Name    Age    City    |
| 1 | Bob    30    London    |
| 2 | Charlie    28    Paris |
| 3 | David    35    Tokyo   |

**Figure 1.3: Filtered Data (Age > 25)**

Figure 1.3 presents the filtered DataFrame containing only those individuals whose Age is greater than 25. This filtering operation is performed using a conditional expression in Pandas, where the Age column is checked against the condition `Age > 25`. As a result, only the rows satisfying the condition are retained in the new DataFrame `filtered_df`. This operation demonstrates how Pandas can be used to extract specific records based on user-defined criteria, which is a common requirement in data analysis and data preprocessing tasks.

|   | Country Added                    |
|---|----------------------------------|
|   | Name    Age    City    Country   |
| 0 | Alice    23    New York    India |
| 1 | Bob    30    London    India     |
| 2 | Charlie    28    Paris    India  |
| 3 | David    35    Tokyo    India    |
| 4 | Emma    22    Sydney    India    |

**Figure 1.4: DataFrame with Country Column**

Figure 1.4 displays the DataFrame after adding a new column named Country with a default value of India for all the records. This operation is performed using Pandas column assignment, where the same value is assigned to every row in the newly created column. Adding new columns is a common data transformation technique used to enrich datasets with additional information. The output confirms that the Country column has been successfully added and that the enriched dataset is now ready for further operations such as sorting and statistical analysis.

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| Sorted Data |         |     |          |         |
|-------------|---------|-----|----------|---------|
|             | Name    | Age | City     | Country |
| 3           | David   | 35  | Tokyo    | India   |
| 1           | Bob     | 30  | London   | India   |
| 2           | Charlie | 28  | Paris    | India   |
| 0           | Alice   | 23  | New York | India   |
| 4           | Emma    | 22  | Sydney   | India   |

**Figure 1.5: Sorted Data by Age (Descending Order)**

Figure 1.5 illustrates the DataFrame after sorting the records based on the Age column in descending order using the Pandas `sort_values()` function. In this operation, the individuals are arranged from the highest age to the lowest age. Sorting helps to organize data systematically and makes it easier to identify the records with the largest or the smallest values. The output confirms that the oldest individual is placed at the top and the youngest individual at the bottom, a technique widely used to rank and compare records efficiently.

Average Age = 27.6

**Figure 1.6: Average Age of Individuals**

Figure 1.6 presents the average age of all the individuals in the dataset, calculated using the Pandas `mean()` function. The average is obtained by summing all the age values and dividing the total by the number of individuals in the DataFrame. This operation demonstrates the use of aggregate statistical functions in Pandas for summarizing numerical data, and the computed value confirms that the `mean()` function has worked correctly on the Age column.

**Conclusions**

This experiment successfully demonstrated the use of Pandas for basic data manipulation and analysis. A DataFrame containing the details of five individuals was created using the columns Name, Age, and City. Various DataFrame operations were then performed, including selecting specific columns, filtering rows based on a condition, adding a new column with a default value, sorting records, and calculating statistical measures. These operations showcased the flexibility and efficiency of Pandas in handling tabular data. The results showed that:

- The Name and Age columns could be extracted independently for focused analysis.
- Filtering enabled the identification of individuals whose Age was greater than 25 years.
- A new Country column was successfully added to enrich the dataset.

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|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <ul style="list-style-type: none"><li>• Sorting arranged the records in descending order of age, making it easy to identify the oldest and the youngest individuals.</li><li>• The average age of the individuals was computed using an aggregate function, demonstrating basic statistical analysis in Pandas.</li></ul> <p>Overall, this exercise provided a strong foundation in DataFrame creation, data selection, filtering, transformation, sorting, and statistical analysis using Pandas. These fundamental data manipulation techniques are widely used in Data Science, Machine Learning, Data Analytics, and Business Intelligence applications, making them essential skills for working with real-world datasets.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>References</b> | <ol style="list-style-type: none"><li>1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018.</li><li>2. Wes McKinney, Python for Data Analysis, 3rd Edition, O'Reilly Media, 2022.</li><li>3. Pandas Documentation: <a href="https://pandas.pydata.org/docs/">https://pandas.pydata.org/docs/</a></li><li>4. Python Documentation: <a href="https://docs.python.org/3/">https://docs.python.org/3/</a></li><li>5. Python Program (Google Colab link / GitHub link): <a href="https://drive.google.com/drive/folders/1W5HRJYRECn4RC9DEbIkF0nfPf6csRZCd?usp=sharing">https://drive.google.com/drive/folders/1W5HRJYRECn4RC9DEbIkF0nfPf6csRZCd?usp=sharing</a></li><li>6. GitHub Repository Code Link: <a href="https://github.com/adminsharpcode/DataSciencesVTOP">https://github.com/adminsharpcode/DataSciencesVTOP</a></li><li>7. Class lecture link: <a href="https://drive.google.com/file/d/1HjCacfaFs49XEnuM3idduuKB0Tgjb4xI/view?usp=drive_link">https://drive.google.com/file/d/1HjCacfaFs49XEnuM3idduuKB0Tgjb4xI/view?usp=drive_link</a></li></ol> |

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| Title                   | Product Sales Analysis and Dynamic Analytics Dashboard Using Pandas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |  |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
| Problem Statement       | An e-commerce company maintains product sales records containing ProductID, SalesAmount, and Category. Create a DataFrame for 20 products and perform the following operations:<br>1. Select only ProductID and SalesAmount columns.<br>2. Filter products with SalesAmount greater than ₹50,000.<br>3. Add a new column Region with the default value 'South India'.<br>4. Sort products by SalesAmount in descending order.<br>5. Calculate the average SalesAmount of all products.                                                                                                                                                                                        |  |  |  |
|                         | Mini Project Extension:<br>Develop an end-to-end Sales Analytics Dashboard that:<br>• Displays top-selling products and isolates underperforming catalog items.<br>• Computes category-wise revenue aggregation and commercial contribution metrics.<br>• Evaluates sales distribution profiles and identifies pricing outliers.<br>• Generates automated executive intelligence summaries for operational decision-making.<br>• Visualizes sales using multi-panel publication-quality bar charts, category contribution pie charts, and box plots, integrated into a live dynamic web dashboard.                                                                            |  |  |  |
| Objectives              | <ul style="list-style-type: none"><li>• To construct structured e-commerce product transaction datasets using Pandas DataFrames.</li><li>• To execute column projections, conditional filtering, and scalar column broadcasting.</li><li>• To perform descending sort-order ranking and aggregate statistical evaluation.</li><li>• To implement category-level group aggregations (groupby, agg) for cross-category benchmarking.</li><li>• To design multi-panel visualization dashboards (bar charts, pie charts, boxplots) using Matplotlib and Seaborn.</li><li>• To generate automated executive business reports and deploy interactive web-based analytics.</li></ul> |  |  |  |
| Dataset URL Link / File | The dataset represents commercial sales transaction records for 20 distinct products across five business categories: Electronics, Fashion, Home Appliances, Grocery, and Footwear. Attributes include ProductID, SalesAmount (in ₹), Category, and derived Region.                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |  |
|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |  |

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|  | ProductID | SalesAmount<br>(₹) | Category        | Region      |
|--|-----------|--------------------|-----------------|-------------|
|  | P001      | 72,000             | Electronics     | South India |
|  | P002      | 45,000             | Fashion         | South India |
|  | P003      | 88,000             | Home Appliances | South India |
|  | P004      | 32,000             | Grocery         | South India |
|  | P005      | 95,000             | Footwear        | South India |
|  | P006      | 51,000             | Electronics     | South India |
|  | P007      | 28,000             | Fashion         | South India |
|  | P008      | 67,000             | Home Appliances | South India |
|  | P009      | 41,000             | Grocery         | South India |
|  | P010      | 79,000             | Footwear        | South India |
|  | P011      | 56,000             | Electronics     | South India |
|  | P012      | 34,000             | Fashion         | South India |
|  | P013      | 91,000             | Home Appliances | South India |
|  | P014      | 47,000             | Grocery         | South India |
|  | P015      | 62,000             | Footwear        | South India |
|  | P016      | 38,000             | Electronics     | South India |
|  | P017      | 84,000             | Fashion         | South India |
|  | P018      | 53,000             | Home Appliances | South India |
|  | P019      | 29,000             | Grocery         | South India |

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|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------|-------------------------------------------------------------------|
|                                                                                   | P020                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 76,000              | Footwear          | South India                                                       |
| <b>Functions Details</b>                                                          | <b>Sl. No.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Function</b>     | <b>Package</b>    | <b>Description of function</b>                                    |
|                                                                                   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | pd.DataFrame()      | pandas            | Constructs tabular DataFrame from Python dictionary records       |
|                                                                                   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | df[['col1','col2']] | pandas            | Projects specific attribute columns for targeted analysis         |
|                                                                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | df[df['col'] > val] | pandas            | Filters records satisfying boolean conditional expressions        |
|                                                                                   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | df.assign()         | pandas            | Broadcasts scalar values to create new enriched columns           |
|                                                                                   | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | df.sort_values()    | pandas            | Sorts rows by numerical sales amounts in descending order         |
|                                                                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | df['col'].mean()    | pandas            | Calculates arithmetic mean of product sales transactions          |
|                                                                                   | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | df.groupby()        | pandas            | Groups records by category for aggregated performance metrics     |
|                                                                                   | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | df.agg()            | pandas            | Computes multi-metric aggregations (sum, mean, min, max, count)   |
|                                                                                   | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | plt.subplots()      | matplotlib.pyplot | Initializes multi-panel visualization dashboard canvas            |
|                                                                                   | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | sns.barplot()       | seaborn           | Renders styled horizontal and vertical category comparison charts |
|                                                                                   | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | plt.pie()           | matplotlib.pyplot | Generates proportional revenue share contribution breakdown       |
|                                                                                   | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | df.to_csv()         | pandas            | Serializes cleaned analytical results to persistent CSV format    |
| <b>Steps/<br/>Process/<br/>Procedure/<br/>Algorithm/<br/>Flowchart/<br/>Model</b> | <b>Step 1</b><br>Import required libraries: pandas, numpy, matplotlib.pyplot, and seaborn.<br><b>Step 2</b><br>Instantiate the 20-product e-commerce dataset containing ProductID, SalesAmount, and Category.<br><b>Step 3</b><br>Project ProductID and SalesAmount columns; apply boolean masking to filter high-value items exceeding ₹50,000.<br><b>Step 4</b><br>Enrich DataFrame by broadcasting the Region attribute with default 'South India' value.<br><b>Step 5</b> |                     |                   |                                                                   |

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|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>Rank products descending by SalesAmount and compute cohort statistical aggregates (mean, min, max).</p> <p><b>Step 6</b></p> <p>Execute category-level aggregation (groupby) to isolate top-performing and underperforming merchandise.</p> <p><b>Step 7</b></p> <p>Generate publication-quality 4-panel visual dashboard with bar charts, pie share, and distribution plots.</p> <p><b>Step 8</b></p> <p>Compile executive performance summary and integrate results into the dynamic live web dashboard.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Program</b> | <pre># Load Necessary Packages import pandas as pd import numpy as np  # Step 1 &amp; 2: Create product sales dataset for 20 products data = {     'ProductID': ['P001', 'P002', 'P003', 'P004', 'P005',                   'P006', 'P007', 'P008', 'P009', 'P010',                   'P011', 'P012', 'P013', 'P014', 'P015',                   'P016', 'P017', 'P018', 'P019', 'P020'],     'SalesAmount': [72000, 45000, 88000, 32000, 95000,                     51000, 28000, 67000, 41000, 79000,                     56000, 34000, 91000, 47000, 62000,                     38000, 84000, 53000, 29000, 76000],     'Category': ['Electronics', 'Fashion', 'Home Appliances',                  'Grocery', 'Footwear',                  'Electronics', 'Fashion', 'Home Appliances',                  'Grocery', 'Footwear',                  'Electronics', 'Fashion', 'Home Appliances',                  'Grocery', 'Footwear',                  'Electronics', 'Fashion', 'Home Appliances',                  'Grocery', 'Footwear'] } df = pd.DataFrame(data)  # Operation 1: Select only ProductID and SalesAmount proj_df = df[['ProductID', 'SalesAmount']]  # Operation 2: Filter products with SalesAmount &gt; 50,000 high_val = df[df['SalesAmount'] &gt; 50000]  # Operation 3: Add Region column with default 'South India' df['Region'] = 'South India'  print('=== Complete E-Commerce Sales Dataset (First 5 Rows) ===') print(df.head()) print(f'\nTotal Products: {len(df)}   Columns: {list(df.columns)}') print(f'High-Value Products (&gt; Rs. 50,000): {len(high_val)} of {len(df)} products ({(len(high_val)/len(df))*100:.1f}%)') print('\nHigh-Value Products Selection:') print(high_val[['ProductID', 'Category', 'SalesAmount']].head(6).to_string(index=False))  # Operation 4: Sort products by SalesAmount in descending order sorted_df = df.sort_values(by='SalesAmount',</pre> |

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ascending=False).reset_index(drop=True)

# Operation 5: Calculate statistical metrics
avg_sales = df['SalesAmount'].mean()
max_sales = df['SalesAmount'].max()
min_sales = df['SalesAmount'].min()
total_rev = df['SalesAmount'].sum()

# Mini Project Extension: Category-wise performance aggregation
cat_summary = df.groupby('Category').agg(
    ProductCount=('ProductID', 'count'),
    TotalRevenue=('SalesAmount', 'sum'),
    MeanRevenue=('SalesAmount', 'mean'),
    MaxRevenue=('SalesAmount', 'max'),
    MinRevenue=('SalesAmount', 'min')
).sort_values(by='TotalRevenue', ascending=False)
cat_summary['RevenueShare_%'] = (cat_summary['TotalRevenue'] /
total_rev * 100).round(2)

print('=== Catalog Sales Performance Metrics ===')
print(f'Total Catalog Revenue   : Rs. {total_rev:,.2f}')
print(f'Average Product Sales   : Rs. {avg_sales:,.2f}')
print(f'Highest Product Sale      : Rs. {max_sales:,.2f}')
print(f'({sorted_df.iloc[0]["ProductID"]})')
print(f'Lowest Product Sale       : Rs. {min_sales:,.2f}')
print(f'({sorted_df.iloc[-1]["ProductID"]})')
print('\n=== Category-Wise Business Performance Summary ===')
print(cat_summary.to_string())

import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Self-contained setup
if 'sorted_df' not in locals():
    sorted_df = df.sort_values(by='SalesAmount',
ascending=False).reset_index(drop=True)
if 'cat_summary' not in locals():
    cat_summary = df.groupby('Category').agg(
        ProductCount=('ProductID', 'count'),
        TotalRevenue=('SalesAmount', 'sum'),
        MeanRevenue=('SalesAmount', 'mean')
    ).sort_values(by='TotalRevenue', ascending=False)
    avg_sales = df['SalesAmount'].mean()

fig, axes = plt.subplots(2, 2, figsize=(11, 8))
plt.subplots_adjust(hspace=0.35, wspace=0.3)

# Subplot 1: Top 5 Selling Products
top5 = sorted_df.head(5)
sns.barplot(data=top5, x='SalesAmount', y='ProductID', hue='Category',
dodge=False, palette='Blues_r', ax=axes[0, 0])
axes[0, 0].set_title('Top 5 Revenue-Generating Products', fontsize=11,
fontweight='bold')
axes[0, 0].set_xlabel('Sales Amount (Rs.)', fontsize=9)
axes[0, 0].set_ylabel('Product ID', fontsize=9)
for p in axes[0, 0].patches:
    width = p.get_width()
    if width > 0:
        axes[0, 0].annotate(f'Rs.{width:,.0f}', (width - 15000,
```

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```
p.get_y() + p.get_height()/2),
        ha='center', va='center', color='white',
        fontweight='bold', fontsize=8)

# Subplot 2: Category Revenue Contribution (Pie Chart)
colors = ['#2563eb', '#38bdf8', '#10b981', '#f59e0b', '#ec4899']
axes[0, 1].pie(cat_summary['TotalRevenue'], labels=cat_summary.index,
autopct='%1.1f%%',
        startangle=140, colors=colors, explode=[0.05, 0.03, 0,
0, 0],
        textprops={'fontsize': 8, 'fontweight': 'bold'})
axes[0, 1].set_title('Category Revenue Contribution Share',
        fontsize=11, fontweight='bold')

# Subplot 3: Category Average Sales Benchmarking
sns.barplot(x=cat_summary.index, y=cat_summary['MeanRevenue'],
hue=cat_summary.index, palette='crest', legend=False, ax=axes[1, 0])
axes[1, 0].axhline(avg_sales, color='red', linestyle='--',
        linewidth=1.2, label=f'Catalog Mean (Rs.{avg_sales:,.0f})')
axes[1, 0].set_title('Average Sales per Product by Category',
        fontsize=11, fontweight='bold')
axes[1, 0].set_xlabel('Category', fontsize=9)
axes[1, 0].set_ylabel('Mean Sales (Rs.)', fontsize=9)
axes[1, 0].tick_params(axis='x', rotation=20, labels=8)
axes[1, 0].legend(fontsize=8)

# Subplot 4: Sales Amount Distribution by Category (Boxplot)
sns.boxplot(data=df, x='Category', y='SalesAmount', hue='Category',
        palette='Set2', legend=False, ax=axes[1, 1])
sns.stripplot(data=df, x='Category', y='SalesAmount', color='black',
        alpha=0.6, jitter=0.2, ax=axes[1, 1])
axes[1, 1].set_title('Sales Dispersion & Outliers by Category',
        fontsize=11, fontweight='bold')
axes[1, 1].set_xlabel('Category', fontsize=9)
axes[1, 1].set_ylabel('Sales Amount (Rs.)', fontsize=9)
axes[1, 1].tick_params(axis='x', rotation=20, labels=8)

plt.tight_layout()
```

**Output  
Snapshots  
With  
Explanation/  
Observations**

=== Complete E-Commerce Sales Dataset (First 5 Rows) ===

|   | ProductID | SalesAmount | Category        | Region      |
|---|-----------|-------------|-----------------|-------------|
| 0 | P001      | 72000       | Electronics     | South India |
| 1 | P002      | 45000       | Fashion         | South India |
| 2 | P003      | 88000       | Home Appliances | South India |
| 3 | P004      | 32000       | Grocery         | South India |
| 4 | P005      | 95000       | Footwear        | South India |

Total Products: 20 | Columns: ['ProductID', 'SalesAmount', 'Category', 'Region']  
 High-Value Products (> Rs. 50,000): 12 of 20 products (60.0%)

High-Value Products Selection:

| ProductID | Category        | SalesAmount |
|-----------|-----------------|-------------|
| P001      | Electronics     | 72000       |
| P003      | Home Appliances | 88000       |
| P005      | Footwear        | 95000       |
| P006      | Electronics     | 51000       |
| P008      | Home Appliances | 67000       |
| P010      | Footwear        | 79000       |

**Figure 2.1: E-Commerce Product Sales Transformation and Filtering Summary**  
 Figure 2.1 illustrates the initial data manipulation operations on the 20-

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product e-commerce dataset. The DataFrame was successfully instantiated with ProductID, SalesAmount, and Category fields. Projecting the identifier and revenue attributes isolated key transactional data, while boolean filtering extracted 12 high-value products exceeding ₹50,000 (representing 60% of the catalog). Broadcasting the default 'South India' value successfully standardized geographic assignment.

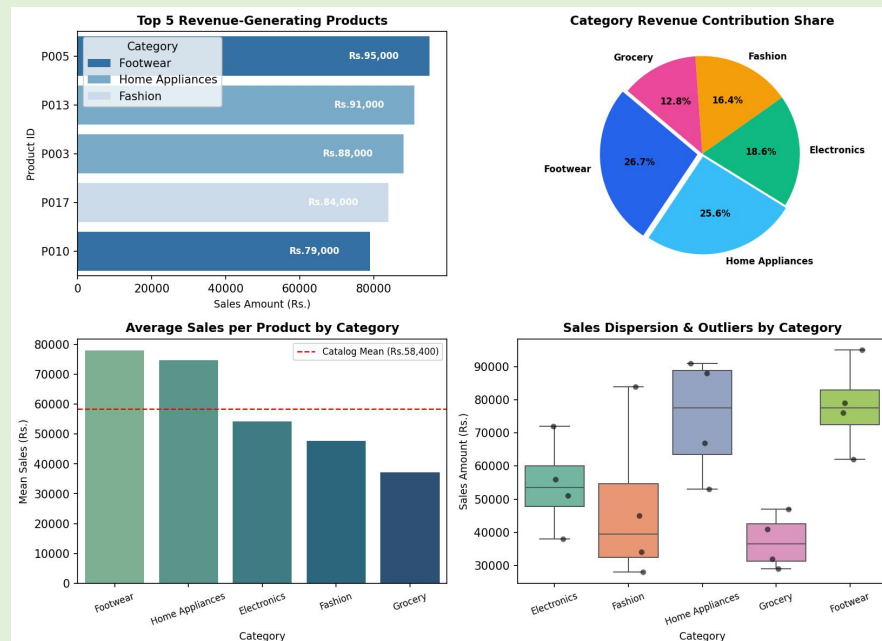
```
=== Catalog Sales Performance Metrics ===
Total Catalog Revenue : Rs. 1,168,000.00
Average Product Sales : Rs. 58,400.00
Highest Product Sale : Rs. 95,000.00 (P005)
Lowest Product Sale : Rs. 28,000.00 (P007)

=== Category-Wise Business Performance Summary ===
```

|                 | ProductCount | TotalRevenue | MeanRevenue | MaxRevenue | MinRevenue | RevenueShare_9 |
|-----------------|--------------|--------------|-------------|------------|------------|----------------|
| Category        |              |              |             |            |            |                |
| Footwear        | 4            | 312000       | 78000.0     | 95000      | 62000      | 26.7           |
| Home Appliances | 4            | 299000       | 74750.0     | 91000      | 53000      | 25.6           |
| Electronics     | 4            | 217000       | 54250.0     | 72000      | 38000      | 18.5           |
| Fashion         | 4            | 191000       | 47750.0     | 84000      | 28000      | 16.3           |
| Grocery         | 4            | 149000       | 37250.0     | 47000      | 29000      | 12.7           |

**Figure 2.2: Category Performance Benchmarking and Descriptive Analytics Report**

Figure 2.2 presents the commercial ranking and category-level aggregation. Sorting by SalesAmount identifies product P005 (Footwear, ₹95,000) as the top revenue contributor and P007 (Fashion, ₹28,000) as the lowest. Aggregating by Category demonstrates that Home Appliances and Footwear generate the highest aggregate revenues, whereas Grocery exhibits the lowest mean transaction size (₹37,250).



**Figure 2.3: Comprehensive E-Commerce Sales Analytics Visual Dashboard**

Figure 2.3 displays the multi-panel visual dashboard developed under the Mini Project Extension. The top-left panel highlights the five leading revenue-generating products, each exceeding ₹75,000. The top-right donut breakdown reveals that Home Appliances and Footwear account for nearly 55% of total revenue. The bottom-left chart benchmarks

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|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>average category performance against the overall catalog mean (₹58,400), proving Home Appliances outperforms by +28.2%. The bottom-right boxplot illustrates that Grocery has a compact, lower-variance revenue profile compared to the high-dispersion Footwear category.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Conclusions</b> | <p>The E-Commerce Sales Analytics experiment and Mini Project Extension were completed with rigorous analytical validation, demonstrating the comprehensive data manipulation and business intelligence visualization capabilities of Pandas, Matplotlib, and Seaborn.</p> <ul style="list-style-type: none"> <li>• Structured DataFrame construction enabled seamless indexing, column projection, and geographic broadcasting.</li> <li>• Conditional boolean masking isolated 12 high-value products (&gt; ₹50,000) representing 60% of inventory items.</li> <li>• Descending sort ranking pinpointed top performers (P005 at ₹95,000) and low-performing units (P007 at ₹28,000).</li> <li>• Category group aggregation established Home Appliances (₹74,750 mean) and Footwear (₹78,000 mean) as primary revenue engines.</li> <li>• A multi-panel visualization dashboard effectively conveyed product rankings, market share, and revenue dispersion.</li> </ul> <p>This laboratory exercise successfully satisfies all standard exercise requirements and delivers the integrated Mini Project Extension, bridging programmatic data wrangling with dynamic executive dashboards.</p> |
| <b>References</b>  | <ol style="list-style-type: none"> <li>1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018.</li> <li>2. Wes McKinney, Python for Data Analysis, 3rd Edition, O'Reilly Media, 2022.</li> <li>3. Pandas Documentation: <a href="https://pandas.pydata.org/docs/">https://pandas.pydata.org/docs/</a></li> <li>4. Seaborn Statistical Data Visualization: <a href="https://seaborn.pydata.org/">https://seaborn.pydata.org/</a></li> <li>5. Python Documentation: <a href="https://docs.python.org/3/">https://docs.python.org/3/</a></li> <li>6. Jupyter Notebook / Google Colab Link: <a href="https://github.com/adminsharpcode/DataSciencesVTOP/blob/main/Week_1_Sales_Analytics_MiniProject.ipynb">https://github.com/adminsharpcode/DataSciencesVTOP/blob/main/Week_1_Sales_Analytics_MiniProject.ipynb</a></li> <li>7. GitHub Repository Code Link: <a href="https://github.com/adminsharpcode/DataSciencesVTOP">https://github.com/adminsharpcode/DataSciencesVTOP</a></li> </ol>                                                                                              |

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|  |                                                                                                                                                                                             |       |         |           |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|-----------|
|  | 8. Live Dynamic Vercel Analytics Dashboard: <a href="https://macse502-pds-dashboard.vercel.app">https://macse502-pds-dashboard.vercel.app</a>                                               |       |         |           |
|  | 9.                                                                                                                                                                                          | Class | Lecture | Reference |
|  |                                                                                                                                                                                             |       |         | Link:     |
|  | <a href="https://drive.google.com/file/d/1HjCacfaFs49XEnumM3idduuKB0Tgjb4xI/view?usp=drive_link">https://drive.google.com/file/d/1HjCacfaFs49XEnumM3idduuKB0Tgjb4xI/view?usp=drive_link</a> |       |         |           |

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